Md Kaif Afran Khan

AI Researcher & Quantum Machine Learning Scientist

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RESEARCH INTERESTS

Quantum Machine Learning (QML), Quantum Key Distribution (QKD), Large Language Models (LLMs), Graph Neural Networks, Time-Series Forecasting, Multimodal Emotion Detection, Data Engineering.

EDUCATION

North South University *Dhaka, Bangladesh*
B.Sc. in Computer Science & Engineering *Expected 2026*
CGPA: 3.62/4.00 | Distinction: Cum Laude | 25% Merit Scholarship

Dhaka Residential Model College *Dhaka, Bangladesh*
Higher Secondary Certificate (HSC) & Secondary School Certificate (SSC)

PUBLICATIONS

Explainable Hybrid ARIMA–ML Forecasting of Bangladesh's GDP: Sectoral Energy Consumption and Carbon Emission Dynamics
5th IEEE International Conference on Robotics, Automation, Artificial-intelligence and Internet-of-Things (2026)

- Developed a Hybrid Residual Learning Ensemble (Hybrid-Stack): ARIMA + ensemble of XGBoost, Random Forest, GBM, SVR on residuals.
- Implemented SHAP explainability. Achieved Best MAPE = 7.60%, outperforming ARIMA and standalone ML baselines.

FSM-LoRA: Dynamic Gradient Gating via Finite State Machines for Energy-Efficient LoRA Fine-Tuning
2nd International Conference on AI and Emerging Technology for Sustainable Future (2026)

- Designed a Finite State Machine controller that dynamically skips redundant backward passes during LoRA fine-tuning.
- Skipped 15–58% of backward passes, achieving up to 58.58% wall-clock time reduction with accuracy within ±0.5% of standard LoRA.
- Evaluated on Phi-2, Qwen2.5-3B, and OpenLLaMA-3B models.

MANUSCRIPTS UNDER REVIEW

Benchmarking Large Language Models Against Classical Machine Learning for Undergraduate Academic Performance Prediction
29th International Conference on Computer and Information Technology

- Compared 7 classical ML models vs. 5 LLMs (zero-shot, few-shot, LoRA) on 872 student records.
- Best ML (Extra Trees): 88.57% accuracy. Best fine-tuned LLM (Gemma-2-9B-IT): 56.57% accuracy.

- Conclusion: Classical ML significantly outperforms current LLMs on structured academic prediction tasks.

TAL-GRN: Topology-Aware Learning Graph Representation Network for Long-Tailed Multi-Class Medical Image Classification

- Utilized adaptive k and learned edge prediction/pruning to enhance F1-score on imbalanced multiclass medical image datasets.

Persona Multiplicity and Deliberative Synthesis in a Single Large Language Model: A Stochastic Multi-Agent Study of Moral Game-Theoretic Choice

- Developed a multi-persona moral reasoning framework generating dynamic personas with independent reasoning and aggregation.
- Outperforms single-shot baselines in stakeholder and consequence coverage metrics.

ONGOING RESEARCH

Quantum Machine Learning for Quantum Key Distribution (QML as QKD)

- Investigating noise types and innovations in BBM92 and E91 quantum key distribution protocols using QML frameworks.
- Exploring entanglement pathways and their impact on secure communication protocols.

Piqagat: A Quantum Adaptive GAT Framework for Muon Regression

- Developing a Quantum Adaptive Graph Attention Network (GAT) framework for muon regression tasks.
- Integrating quantum circuit architectures with classical graph attention mechanisms to enhance predictive performance.

RESEARCH & PROFESSIONAL EXPERIENCE

AI Engineer *The Data Island* *2025*

- Built an automated form-filling system for Aselo company, extracting structured form data from S3 transcripts using AWS Bedrock (Claude).
- Tech Stack: FastAPI, AWS S3, AWS Bedrock, React + TypeScript.
- Implemented MFA session support, real-time chat view, and scalable architecture ready for Lambda/ECS deployment.

Core Volunteer & Currency Tech Lead *Farzana Afroz Foundation (NGO) 2022 – Present*

- Lead technical development for the organization's digital platforms and currency-related technology initiatives.
- Maintained and scaled the foundation's official website (farhanaafrozfoundation.org).

SELECTED TECHNICAL PROJECTS

MarketPlus – Local Stock Analytics Dashboard (Data Engineering)
AWS S3, Athena, Glue, Spark, Prefect, FastAPI, React

- Engineered a medallion architecture (Bronze → Silver → Gold) data pipeline for stock market analytics.
- Cloud ingestion via EventBridge + Lambda (Alpha Vantage API), with local Spark processing orchestrated by Prefect.
- Computed daily returns, moving averages, volatility, and top movers for real-time dashboard visualization.

STORE – Multi-Tenant Smart Multi-Store ERP
FastAPI, MongoDB, React, React Native, Redis, Docker

- Built a full ERP platform with three surfaces: Backend API, Web Dashboard, and Mobile Worker App + AI Assistant.
- Modules included Auth/RBAC, Inventory, Sales, Attendance, Analytics, and AI Assistant (OpenRouter) with multi-tenancy enforced via businessId.

Denz E-Commerce (Live: denzapparel.com)
Next.js 14, TypeScript, Supabase, Tailwind

- Modern full-stack e-commerce platform with Google OAuth, admin dashboard, and secure role-based access.
- Implemented HttpOnly cookies, Row Level Security (RLS), and Cloudinary integration.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript/JavaScript, PHP, SQL, C++
Quantum Machine Learning: Qiskit, PennyLane, JAX, Cirq, Quantum Neural Networks (QNNs), Variational Quantum Circuits (VQCs), Quantum Key Distribution (QKD)
AI/ML: Scikit-learn, XGBoost, SHAP, LoRA Fine-tuning, LLMs (Prompting & Fine-tuning), AWS Bedrock, Graph Neural Networks (GNNs)
Data Engineering: Kafka, Spark Structured Streaming, dbt, Airflow, Prefect, Medallion Architecture, AWS Glue, AWS Athena
Backend & Cloud: FastAPI, Flask, Node.js, AWS (S3, Lambda, EventBridge, Glue, Athena, ECS/Fargate, EC2), Docker, Redis
Frontend: React, Next.js, Tailwind, Bootstrap, React Native (Expo)
Databases: MongoDB, PostgreSQL, MySQL, Supabase, Sanity CMS

CERTIFICATIONS & AWARDS

- AWS Data Engineer Practitioner – Coursera
- Full Stack Developer (MERN Stack) – Ostaad Platform
- Data Science Short Course – Jennifer Widom (Stanford Online)
- 25% Merit Scholarship – North South University
- Cum Laude Distinction – North South University

LANGUAGES

English (IELTS 7.5), Bangla (Native)